CAPSTONE PROJECT REPORT

Agentic AI

HR Assist

AI-Powered Company HR Policy Bot

Domain: Human Resources | Enterprise Chatbot | RAG + Agentic AI

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1. Problem Statement

In most organisations, employees spend significant time seeking answers to routine HR policy questions — questions about leave entitlements, payroll cycles, reimbursement limits, resignation notice periods, and benefits eligibility. These queries typically require the employee to email the HR department, search through policy PDFs, or wait for a response that could take days. The HR team, in turn, spends a disproportionate amount of time answering repetitive queries that could be automated, diverting attention from high-value HR functions such as talent management, engagement, and compliance.

The core problem is a gap between policy information (which exists, but is siloed in documents) and policy access (which is slow, inconsistent, and human-dependent). Employees either do not know what their entitlements are or cannot quickly find the authoritative answer. This leads to policy violations due to misunderstanding, employee dissatisfaction from delayed responses, and avoidable HR workload.

Additionally, existing FAQ pages and static policy portals are not conversational — they cannot handle follow-up questions, personalise responses to context (e.g., "I joined 3 months ago, am I eligible for WFH?"), or perform calculations (e.g., "If I resign today, when is my last working day?"). There is a clear need for a 24/7, intelligent, context-aware HR policy assistant.

2. Solution — HR Assist

HR Assist is a production-grade, agentic AI chatbot built on LangGraph, ChromaDB, and the Groq-hosted LLaMA 3.3 70B model. It provides employees with instant, accurate, and grounded answers to HR policy questions via a conversational Streamlit interface. The agent goes beyond simple retrieval-augmented generation (RAG) by incorporating intelligent routing, tool use, faithfulness self-evaluation, and multi-turn memory.

2.1 Agent Pipeline — How It Works

Every employee query passes through an eight-node LangGraph StateGraph pipeline:

Figure 1 — LangGraph StateGraph pipeline (left to right: memory → router → retrieve/skip/tool → answer → eval → save)

- **Memory:** memory node updates the rolling conversation window (last 6 messages) and extracts the employee's name if introduced.
- **Router:** router node classifies the query into one of three paths: retrieve (semantic search), memory_only (conversation follow-up), or tool (date/number calculation).
- **Retrieve:** retrieve node performs vector similarity search over 12 HR policy documents in ChromaDB, returning the top-3 most relevant chunks with their topic labels.

- **Tool:** tool node runs a Python datetime-based leave calculator — computing accrued Annual Leave, Casual Leave, notice period end date, and remaining business days in the year.
- **Answer:** answer node constructs a grounded response using only the retrieved context, with an HR escalation detector that flags queries requiring human HR intervention.
- **Eval:** eval node acts as an LLM-as-judge, scoring the answer's faithfulness to the retrieved context on a 0.0–1.0 scale. If the score falls below 0.7, the answer node is retried (up to 2 times).
- **Save:** save node appends the final answer to the conversation history for memory continuity.

3. Key Features

3.1 Mandatory Capability Checklist

#	Capability	Implementation
1	**LangGraph StateGraph (3+ nodes)**	8 nodes: memory → router → retrieve/skip/tool → answer → eval → save
2	**ChromaDB RAG (10+ documents)**	12 HR policy docs embedded with MiniLM-L6-v2, top-3 retrieval
3	**Conversation Memory**	MemorySaver + thread_id; rolling 6-message window
4	**Self-Reflection / Eval Loop**	LLM-as-judge faithfulness scoring; retries up to 2× if score < 0.7
5	**Tool Use**	Leave calculator: accrued days, notice period end date, working days left
6	**Deployment**	Streamlit UI with real-time chat, sidebar stats, escalation badges

Table 1 — All 6 mandatory capstone capabilities and their implementation in HR Assist

3.2 Knowledge Base — 12 HR Policy Documents

The ChromaDB vector store contains 12 authoritative HR policy documents covering every major topic an employee might enquire about:

- Annual Leave (18 days/year, carry-forward rules, half-day leave)
- Sick Leave (12 days/year, hospitalisation leave, medical certificate rules)
- Work From Home (hybrid model, eligibility, Monday/Friday restrictions)
- Payroll & Salary (disbursement dates, CTC structure, TDS, salary advances)
- Reimbursements (travel, meals, km-based vehicle rates, approval workflow)
- Performance Appraisal (rating scale 1–5, increment bands, PIP process)
- Onboarding & Probation (6-month probation, confirmation process, leave eligibility)
- Resignation & Notice Period (60-day notice, buy-out, garden leave, absconding)
- Employee Benefits (group mediclaim, PF, gratuity formula, life insurance, EAP)

- Code of Conduct & Disciplinary Process (4-step process, severe misconduct)
- POSH Policy (ICC constitution, complaint process, retaliation protection)
- Public Holidays & Company Calendar (12 holidays, restricted holidays, year-end shutdown)

3.3 Streamlit UI — Employee-Facing Interface

The Streamlit front-end is a two-column layout providing a full chat experience with real-time metadata transparency:

- Live faithfulness score, routing decision, and source document displayed per response.
- Colour-coded average faithfulness indicator: green (≥0.80), yellow (≥0.60), red (<0.60).
- Six quick-prompt buttons in the sidebar for the most common HR queries.
- HR escalation badge (red warning) when the response recommends human HR contact.
- Session management: unique thread IDs, query counter, and one-click conversation reset.

HR Assist — Company Policy Bot
Ask about leave, payroll, reimbursements, appraisals, benefits, resignation, and more.

You: How many annual leave days do I get per year?

HR Assist: Every confirmed employee is entitled to 18 days of Annual Leave (AL) per calendar year, accrued at 1.5 days per month. AL can be carried forward up to 9 days into the next year; any balance above 9 days lapses on December 31st.
Faithfulness: 0.97 | Route: retrieve | Source: Annual and Casual Leave Policy

12 policy docs loaded | Session: a3f7c2b1 | Queries: 3 | Avg Faithfulness: 0.94

Figure 2 — Simulated HR Assist chat interface showing a policy query, grounded answer, and metadata footer

4. Technology Stack

Layer	Technology	Purpose
LLM	Groq · LLaMA 3.3 70B	Fast inference for routing, answering, eval
Orchestration	LangGraph (StateGraph)	Multi-node agentic pipeline
Memory	LangGraph MemorySaver	Persistent multi-turn conversation
Vector Store	ChromaDB (in-memory)	Semantic document retrieval
Embeddings	all-MiniLM-L6-v2	Sentence-level encoding of KB docs
LLM SDK	LangChain / langchain-groq	LLM message abstraction

Layer	Technology	Purpose
Tool	Python datetime module	Leave calculator & date arithmetic
UI Framework	Streamlit	Chat-based employee interface
Environment	python-dotenv	Secure API key management

Table 2 — Full technology stack with layer, component, and purpose

The architecture is designed to be lightweight and fast. Groq's LPU inference engine delivers sub-second LLM responses, making the agentic pipeline feel responsive to employees. ChromaDB runs entirely in-memory, eliminating the need for external database infrastructure. The entire application is deployable with a single command (streamlit run capstone_streamlit.py) from a machine with Python 3.10+ and a valid GROQ_API_KEY.

5. Unique Points & Design Decisions

5.1 LLM-as-Judge Self-Evaluation Loop

Rather than relying on RAGAS or an external evaluation framework (which would require an OpenAI API key), HR Assist uses the same LLaMA 3.3 70B model to evaluate its own answers. The eval node scores faithfulness on a 0–1 scale and automatically retries the answer node if the score falls below 0.70. This creates a self-correcting loop that improves answer quality without human intervention and adds measurable quality assurance directly into the agent pipeline.

5.2 Intelligent Three-Way Router

Most RAG systems send every query to the vector store, wasting tokens and latency on follow-up questions like "What did you just say?" or "Can you summarise that?". HR Assist's router classifies queries into retrieve, memory_only, or tool — skipping retrieval entirely for conversational follow-ups and routing calculation queries directly to the Python tool. This design reduces unnecessary LLM calls and improves response speed.

5.3 Domain-Specific Leave Calculator Tool

The built-in leave calculator is not a generic arithmetic tool — it is calibrated to the company's specific HR rules. It computes accrued Annual Leave at 1.5 days/month, accrued Casual Leave at 0.5 days/month, notice period end dates (60 calendar days from today), and remaining business days in the calendar year. This directly answers one of the most common and error-prone calculations employees attempt manually.

5.4 Automatic HR Escalation Detection

The answer node inspects its own output for escalation signals (phrases such as "contact HR", "PIP", "ICC", "disciplinary"). When detected, a red warning badge is displayed in the UI directing the employee to hr@company.com or the HR portal. This ensures the bot gracefully hands off complex or sensitive cases rather than attempting to resolve them autonomously.

5.5 Policy-Faithful Design — No Training Data Leakage

The system prompt explicitly instructs the model to answer using only the retrieved context and never draw on its training data. If a query falls outside the 12 policy documents, the agent responds with a specific message directing the user to HR, rather than confabulating an answer. This is a critical trust-building design decision for an enterprise HR application where incorrect policy information could have legal or financial consequences.

6. Future Improvements

6.1 Personalised Leave Balance Tracking

Currently, the leave calculator computes theoretical accruals based on the current date without knowledge of how many leave days an employee has already taken. Integrating with an HRMS (e.g., Darwinbox, BambooHR) via API would allow the agent to retrieve each employee's actual leave balance, taken days, and pending approvals — making the calculator answers precise rather than approximate.

6.2 Persistent Vector Store and Document Management

The ChromaDB collection is rebuilt in-memory on every application start. Moving to a persistent ChromaDB instance (or a managed vector store like Pinecone or Weaviate) would allow the HR team to add, update, or remove policy documents without redeploying the application. A simple HR-facing admin UI could allow document uploads that are automatically chunked, embedded, and indexed.

6.3 Multi-Language Support

Many employees in organisations across India may prefer to ask questions in Hindi or regional languages. Integrating a translation layer (or using a multilingual embedding model such as paraphrase-multilingual-MiniLM-L12-v2) would extend the bot's reach to non-English-speaking employees without requiring new policy documents.

6.4 HR Portal Integration & Ticket Raising

When the agent detects an escalation-worthy query, it currently directs the employee to contact HR manually. A future version could integrate with the company's HR ticketing system (e.g., Freshdesk, Jira Service Management) to automatically raise a ticket pre-populated with the employee's question, the bot's response, and the relevant policy section — closing the loop without requiring the employee to repeat information.

6.5 Voice Interface

Adding speech-to-text (e.g., OpenAI Whisper) and text-to-speech (e.g., ElevenLabs) would enable a voice-based HR assistant accessible from mobile or desk phones — useful for employees on the factory floor or in field roles who cannot easily type a query.

6.6 Analytics Dashboard for HR Teams

Logging all queries, routes, faithfulness scores, and escalation flags to a database would enable an HR analytics dashboard showing: most frequently asked policy topics, average faithfulness trends, escalation rates by department, and queries the bot failed to answer confidently. This data would directly inform which policy documents need to be updated or expanded.

Conclusion

HR Assist demonstrates that a production-grade, domain-specific AI agent can be built rapidly with modern open-source tooling. By combining LangGraph's stateful orchestration, ChromaDB's semantic retrieval, Groq's fast inference, and a self-evaluation loop, the system reliably answers the most common HR policy queries with measurable faithfulness. The project satisfies all six mandatory capstone capabilities and introduces several domain-specific design decisions — the three-way router, the policy-faithful prompt, the automatic escalation detector, and the domain-calibrated leave calculator — that make it meaningfully more useful than a generic RAG chatbot. The improvements outlined in Section 6 provide a clear roadmap toward a fully production-deployed, HRMS-integrated AI assistant.